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Answer Key

- 1 **B – Substance B**
- 2 Liquid particles are **close together** with small spaces between them. They are **randomly arranged** and can **move/slide past one another**.
- 3 Gases can be compressed because their particles are **far apart with large spaces between them**. Liquid particles are already **close together**, so liquids are difficult to compress.
- 4 The change is **deposition**. Gas particles **lose thermal energy**, move more slowly and become **closely packed in fixed positions** to form a solid.
- 5 **Option B – insulated container**. It reduces the transfer of **thermal energy** from the surroundings to the ice, so the ice melts more slowly.
- 6 **B – deep inside the freezer**. The lower, more constant temperature causes the juice to **lose thermal energy faster**, so it freezes more quickly.
- 7 When the freezer door is open, **thermal energy is transferred** from the warmer surroundings to the frozen food. Its particles **gain energy and move/vibrate faster**, causing melting. **Solution:** keep the freezer door closed.
- 8a **Deposition:** water vapour changes directly from **gas to solid**. Particles **lose thermal energy**, slow down and become closely packed in fixed positions.
- 8 Cover the plants with a **frost cloth/cover**. It reduces
- b **heat/thermal energy loss** from the plants and surrounding air, helping prevent frost formation.
- 9 The statement is **incorrect**. Substance B is a **liquid** because it takes the shape of its container but **keeps a fixed volume**.

- 10** The conclusion is **supported**. Ice on the sunny windowsill melted in **12 min**, compared with **28 min** in the shade. More thermal energy causes particles to gain energy, so melting occurs faster.

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- 11** The conclusion is **supported**. Container A (small surface area) had **45 mL** remaining, while C (large surface area) had only **29 mL** remaining. Therefore, more water evaporated from the container with the larger surface area. A larger surface area exposes **more particles at the surface**, so more particles have enough energy to **escape into the gas state**, increasing the rate of evaporation.
- 12** The conclusion is **supported**. At **15°C**, ice took **22 min** to melt; at **25°C**, **14 min**; and at **35°C**, only **7 min**. As temperature increases, ice particles **gain thermal energy** and **vibrate faster** until they overcome the forces holding them in fixed positions and the solid melts. The evidence is **sufficient to support the conclusion within the tested temperatures**, because equal-sized ice cubes were used and a clear pattern is shown.